MIL-C-23044A(MC)
4 August 1967
SUPERSEDING
MIL-C-23044 (MC)
9 October 1961

MILITARY SPECIFICATION

CHEST, COLLAPSIBLE TANK

1. SCOPE

1.1 This specification covers one type skid mounted aluminum alloy chest.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-200/3	- Aluminum Alloy Bar,Rod,Shapes, and Tubes, Extruded 2024.
QQ-A-200/8	- Aluminum Alloy Bar, Rod, Shapes, and Tube, Extruded, 6061.
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet.
QQ-A-601	- Aluminum-Alloy Sand Castings.
QQ-P-416	- Plating, Cadmium (Electroposited).
QQ-S-781	- Steel, Strapping, Flat.
QQ-W-470	- Wire, Steel, Carbon, Spring, Music.
TT-E-527	- Enamel, Alkyd, Lusterless.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-R-1149	- Rubber Sheets, Strips, and Gaskets, Solid, Synthetic, Medium and Medium Hard.
MIL-W-8604	- Welding of Aluminum Alloys: Process for.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.

FSC 4930

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MIL-STD-130 Identification Marking of US Military Property.
MS-33586 Metals, Definition of Dissimilar.

DRAWINGS

MARINE CORPS

MCD-5136-5-1 - Chest, Aluminum, Tank.
MCD-5136-5-l-l-Details, Chest, Aluminum, Tank.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards. for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Preproduction sample. As soon as practicable after award of the contract or order, and prior to submission of any complete item for acceptance, the contractor shall furnish a preproduction sample for determination of conformance to this specification. The preproduction sample shall be representative of the product which the manufacturer will subsequently produce in fulfillment of the contract. Approval of the preproduction sample by the activity concerned shall not relieve the contractor of the obligation to supply chests conforming to this specification. Any changes or deviations from the preproduction sample shall be subject to approval of the contracting officer.

3.2 Material. Material shall be as specified herein and on the applicable drawings. Material not definitely specified shall be such as to enable the chest to meet the requirements of this specification.

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3.2.1 Metals. Metal parts shall be either corrosion-resistant or shall be protected against corrosion by means of chemical or electrolytic finishes and treatments, or a combination thereof. Steal parts shall be cadmium plated. All treatments and finishes shall be in accordance with applicable specifications.

3.2.1.1 Dissimilar metals. Contact between dissimilar metals shall be avoided. Where it is necessary that dissimilar metals be assembled in intimate contact with each other, an interposing, insulating or mutually compatible material shall be used. Dissimilar metals are defined in MS-33586.

3.2.1.2 Aluminum alloy sand castings. Aluminum alloy sand castings shall conform to QQ-A-601. Castings shall be of high quality, clean, sound and free from porosity, blow holes, cracks, and other production defects.

3.2.1.3 Plate and sheet aluminum alloy. Plate and sheet aluminum alloy for the chest sides, ends, lid and bottom shall conform to QQ-A-250/11, class T6 and shall not be less than 0.090 inch thickness.

3.2.1.4 Aluminum bars, rods and shapes Aluminum bars, rods and shapes, extrusions for the lid packing gland, stiffeners, corner posts and skids shall conform to QQ-A-200/8, class T6, or QQ-A-200/3, class T6. The extrusion shall have rounded edges with material thickness not less than 0.090 inch. The carrying bars shall be not less than 0.075 inch diameter.

3.2.1.5 Springs. Springs shall be steel wire conforming to QQ-W-470 and shall be cadmium plated in accordance with QQ-P-416, type II, class 1.

3.2.2 Fungus inert material. Unless otherwise specified, materials which provide a nutrient medium for fungus and insects shall not be used.

3.2.3 Military standard (MS) Parts. MS parts shall be used wherever they are suitable for the purpose in accordance with applicable drawings and specifications.

3.2.4 Gasket material. Gasket material shall conform to MIL-R-1149, type I, class 2.

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3.3 Finish.

3.3.1 Painting. The exterior and interior of chest, . including the lid, shall be treated and painted in accordance with MIL-T-704, type B,except that the finish coat of paint shall conform to TT-E-527. Color shall match Marine Corps green No. 23 lusterless (see 6.5).

3.3.2 Anodizing. Aluminum parts not painted, shall-be color anodized in accordance with MIL-A-8625; the color shall match Marine Corps green No. 23, lusterless (see 6.5).

3.4 Design and construction.

3.4.1 Weight. The weight of each completely assembled chest with lid, when empty, shall not exceed 325 pounds and shall be kept to the minimum consistent with the requirements of this specification.

3.4.2 Lifting and towing hardware. Four towing eyes, two on each end, and four lifting hooks, two on each side, shall be provided. Lifting hooks shall have the ends welded together to prevent spreading. Lifting hooks and towing eyes shall be fabricated and located as shown on Drawing No. 5136-5-1.

3.4.3 Carrying bars. Each chest shall be provided with a carrying bar on each side as shown on Drawing No. 5136-5-1 (see 3.2.1.4).

3.4.4 Lid fastening devices. Seven latches shall be provided on one side of the chest. The latches shall be spring loaded, quick acting, dog type, and shall be self positioning along the chest side when not in use. Seven hold-down devices, attached to the side of the chest opposite the latches, shall provide a positive hold for the lid in the longitudinal, transverse, and vertical directions. The hold-down devices shall permit rapid removal of the lid, and shall function when the lid is installed in any - given mating position and then turned end for end (180° rota ion in the horizontal plane). The latches shall not accidenta engage as a result of any of the conditions specified herein.

3.4.5 Lid. The lid shall be constructed as shown on the applicable drawings and as specified herein. When installed on the chest by means of the fastening devices, the lid, initially and after any of the conditions specified herein.

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shall provide an effective seal sufficient to protect the chest contents during storage, handling and shipping, when inspected as specified in 4.6.2(a).

3.4.6 Storage compartment. A 9-inch-wide compartment, for storing tank fittings and the tank repair kit, shall be provided at one end of the chest.

3.4.7 Skids. Two skids, not less than 2-1/2 inches wide shall be provided on the bottom of each chest. The skids shall run the entire length of the chest and shall be turned up at each end a minimum of 3 inches on a 14- to 16-inch radius. The skid webs shall be slotted to permit handling the chest by forklift truck.

3.4.8 Stacking lugs. Four stacking lugs shall be provided on the top of each chest lid for use as a stacking aid.

3.4.9 Welding. The quality and strength of all welds shall be such as to enable the chests to meet the requirements of this specification.

3.4.9.1 Aluminum. All aluminum welding shall be by the inert gas shielded arc method in accordance with MIL-W-8604.

3.4.10 Interchangeability. All parts having the same manufacturer's part numbers shall be physically and functionally interchangeable.

3.4.11 Special tools. No special tools shall be required for operation and maintenance of the chests or the components thereof.

3.4.12 Screw threads. All screw threads shall be in accordance with H-28.

3.4..13 Identification marking. Marking for identification shall be in accordance with the drawings and MIL-STD-130.

3.4.14 Nameplate. A "US" nameplate shall be securely attached in a conspicuous location, to each assembled unit. For the Marine Corps, the plates will be furnished upon request to the Commandant of the Marine Corps, Code CSG, Headquarters U. S. Marine Corps, Washington, D. C. 20380. The contractor shall legibly imprint the required data in the blank spaces as directed by the procuring activity (see 6.2).

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3.4.15 Cleaning All parts shall be clean and free from lint, sand, metal chips and weld splashes.

3.4.16 Interior. The chest interior shall be free from all sharp protrusions, corners, burrs, etc., that might damage the fabric tank (see 6.1).

3.5 Performance and product characteristics.

3.5.1 Lid operation. Initially, and after any of the conditions specified herein, it shall be possible to install and to remove the chestlid by normal procedures. With the lid lugs engaged by the hold-down devices, and the latches disengaged, it shall be possible to raise the lid and disengage it from the chest body.

3.5.1.1 End for end rotation. The lid shall be capable of rotating 180° in addition to the requirements of 3.5.1.

3.5.2 Lifting. The chest, with the lid latched in place shall be capable of meeting the test specified in 4.6.3 without permanent deformation or damage to the lifting hooks, the chest, or the lid, and without any latches becoming disengaged.

3.5.2.1 Bar lift. The chest, with the lid latched in place shall be capable of meeting the test specified in 4.6.3.1 without permanent deformation or damage to the bar, the chest, or the lid, and without any latches becoming disengaged.

3.6 Service conditions.

3.6.1 Drop. With the lid latched in place, the chest shall be capable of withstanding the drop tests specified in 4.6.4. and 4.6.4.1 without being damaged beyond use and without any latches becoming disengaged. Repairable, superficial damage is to be expected and will be permitted. After the drop test the chest shall be capable of meeting the performance and product characteristics specified in 3.5.

3.6.2 Towing. With the lid latched in place (see 3.4.4) the chest shall be capable of meeting the requirements of 4.6.5 without being damaged beyond use. During towing, no latches shall become disengaged. Minor repairable damage will be permitted. After towing, the chest shall be capable of meeting the performance and product characteristics specified in 3.5.

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3.7 Workmanship. Chests shall be manufactured in a careful and workmanlike manner in accordance with sound practice consistent with the current state of the art.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Test conditions. Unless otherwise specified, all tests specified herein may be performed under any natural combination of prevailing ambient conditions.

4.3 Classification of tests. The inspection and testing of chests shall be classified as follows:

(a) Preproduction sample tests (see 4.4 and 6.2).
(b) Acceptance tests (see 4.5).

4.4 Preproduction sample tests. A completely fabricated chest shall be subjected to all the inspections and tests specified in 4.6.1 through 4.6.5. These tests shall be made by the contractor in the presence of a Government representative (see 6.3) prior to any acceptance by the Government. These tests are usually performed once during the life of a contract and need not be repeated unless a change is made in design, material or methods of construction.

4.5 Acceptance tests. Unless otherwise specified, each chest submitted for acceptance shall be subjected to the inspections and tests specified herein, in the order shown. These tests shall be performed by the contractor in the presence of a Government representative (see 6.3).

4.6 Test Procedures. Following are detailed examinations and test procedures by which conformance to the product requirements of this specification shall be determined. Test procedures for the constituent parts or materials for the

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chest and for the packaging thereof, that are covered in documents referenced in the basic specifications shall not be repeated (see 4.1).

4.6.1 Visual and mechanical inspection. Chests shall be inspected to verify that the materials, design, construction, physical dimensions, weight, marking and workmanship are in accordance with the requirements of this specification, and Drawing Numbers 5136-5-1 and 5136-5-1-1.

4.6.2 Lid operation test. Lid operation shall be tested as follows:

(a) Install lid and latch into place. Visually inspect gasket for effective seal.

(b) Unlatch lid and remove from chest.

(c) Turn lid end for end (180° rotation in a horizontal plane), assemble with the chest and inspect for effective seal.

(d) Repeat (a) through (c), inclusive for a total of 25 cycles for preproduction testing; or a total of 5 cycles for acceptance testing.

(e) With the lid lugs engaged by the hold-down devices and the latches disengaged, raise the lid to a 45° angle and allow it to fall to the chest. This test shall be repeated for a total of five times. Inspect the lid for structural failure and loosened gasket.

4.6.3 Lifting. The chest, loaded with sand or other approved ballast to an evenly distributed gross weight of 2,600 pounds minimum, and with the lid latched in place, shall be hoisted so that the entire weight of the chest and its contents are supported by the four lifting hooks (see 3.4.2) for a minimum period of 10 minutes. During and after the test, the lifting hooks, the chest and the lid-shall be inspected for evidence of permanent deformation or failure, and the latches shall be observed for malfunction. After this test, lid operation shall be tested as specified for acceptance testing in 4.6.2.

4.6.3.1 Bar lift. With the chest loaded with sand or other approved ballast to an evenly distributed gross weight

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of 1,200 pounds minimum and the lid latched in place, an upward pull of 300 pounds minimum shall be exerted on either bar (see 3.4.3) selected at random. During and after the test, the bar, the chest and the lid shall be inspected for evidence of permanent deformation or failure, and the latches shall be observed for malfunction. After this test, lid operation shall be tested as specified for acceptance testing in 4.6.2.

4.6.4 Drop. The chest, loaded with sand or other approved ballast to an evenly distributed gross weight of 1,200 pounds minimum, and with the lid latched in place, shall be suspended by the lifting hooks, with the bottom surface of the skids a minimum of 2 feet above a concrete floor or equivalent. The chest shall be released by means of a suitable triggering device and shall fall free to the concrete surface. After the test the lifting hooks, the chest, and the lids shall be inspected for evidence of permanent deformation or failure and the latches shall be examined for malfunction. After this test, lid operation shall be tested as specified for acceptance testing in 4.6.2 and the chest shall be subjected to the lifting tests specified in 4.6.3 and 4.6.3.1.

4.6.4.1 End drop. The chest shall be tested as specified in 4.6.4 except that it shall be raised to a level where the bottom surface of the skids are at a minimum height of 6 feet above the concrete surface. One end of the chest shall be supported on a bulkhead and the free end allowed to fall to the concrete surface. After this test, the chest and lid shall be examined for evidence of permanent deformation or failure, and the latches shall be examined for malfunction. Lid operation shall then be tested as specified in 4.6.2 for acceptance testing.

4.6.5 Towing. With chest in the initial condition specified in 4.6.4, it shall be towed from either end (selected at random) by means of the towing eyes for a minimum distance of 400 feet across a concrete pavement at a speed of 5 to 7 miles per hour. During towing, the latches shall be observed for malfunction. After towing, the chest shall be subjected to the following examinations and tests:

(a) Visual examinations for ‘unrepairable damage-
(b) Lid operation as specified for acceptance testing in 4.6.2.
(c) Lifting as specified in 4.6.3 and 4.6.3.1.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. Each Chest covered by this specification becomes its own package and requires no preservation.

5.1.1 Strapping. After the lid is secured, the chest shall be banded with at least one steel strap conforming to QQ-S-781, minimum width 1-1/2 inches. Wooden blocks shall be provided along the edges of the lid of the chest to prevent the strapping from damaging the chest or lid. The steel straps shall be stapled to the protective wooden blocks.

5,2 Packing. Not applicable.

5.3 Marking. In addition to any special marking required by the contract or order each chest shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The chests are intended for use as a storage, handling and shipping container for collapsible fuel tanks, ground cloths, fittings and repair kit which form a part of the amphibious assault fuel system.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Whether a preproduction sample is required, and if so, details of testing, if other than those specified herein (see 3.1).

(c) Nameplates required and nameplate data (see 3.4.14).

6,3 Preproduction sample. The preproduction sample shall be examined and tested for approval at the contractor's plant or at an independent commercial laboratory acceptable to the procuring activity. The preproduction sample tests will be witnessed by a representative of the procuring activity. Approval of the preproduction sample shall be by the contracting officer.

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6.4 Notification. Unless otherwise specified, the contractor shall notify the procuring agency 15 calender days in advance of any testing requiring the presence of a Government representative.

6.5 Color. Color chips for Marine Corps green No. 23 (lusterless), may be obtained from the Commandant of the Marine Corps (Code CSY-10), Headquarters, U. S. Marine Corps, Washington, D. C. 20380.

6.6 Marginal notations. Asterisks are not used in the revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - MC

Project Number:
4930-N005

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH OBSOLETE

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Headquarters U. S. Marine Corps
Washington, D. C. 20380

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Commandant of the Marine Corps (Code CSY-10)
Headquarters Marine Corps
Washington, D. C. 20380

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